

Daily Reflection Tree

A Data Science Analysis of Daily Reflections, Mood, and Productivity

Prepared by Subhash Uniyal | Python · Pandas · NLTK · scikit-learn · Hugging Face Transformers

Executive Summary

This project applies a complete data science workflow — cleaning, exploratory analysis, NLP, sentiment and emotion analysis, topic modeling, and machine learning — to 517 daily reflection records. Mood, energy, and productivity were found to be moderately-to-strongly correlated with each other ($r \approx 0.60$ – 0.62), and the reflection text honestly agrees with the numeric ratings (positive-word count correlates $+0.47$ with mood, negative-word count -0.46). A machine learning model was built to predict next-day productivity using a proper chronological train/test split; it honestly did not outperform a simple mean-prediction baseline, a real and defensible finding given the dataset size, reported transparently rather than adjusted to look stronger.

Problem Statement

Turning free-form daily writing into structured insight is a genuine unstructured-to-structured data problem. Each day produces text (a reflection) alongside self-reported ratings (mood, energy, productivity) and free-form tags. This project asks: how does sentiment change over time and does it agree with self-reported mood; which topics come up most often and how do they relate to mood and productivity; and can prior days help predict tomorrow's productivity?

Dataset

517 synthetic daily reflection records spanning January 2025 to August 2026. The dataset is synthetic by design — generated rather than collected from a real person's private journal — but built with realistic structure: mood, energy, and productivity are drawn from a shared “day quality” latent factor plus independent noise, there is a mild weekend productivity dip, a small day-to-day autocorrelation, and roughly 8% of days have no entry, mirroring real journaling gaps.

Methodology

Data Cleaning

- Checked for missing values, duplicate rows, duplicate reflection text, invalid dates, out-of-range ratings, and empty text
- Built a negation-safe text cleaning pipeline — raw text is always preserved separately from cleaned text, and negation words (not, never, n't) are deliberately excluded from stopwords removal so sentiment signal is not destroyed

Exploratory Data Analysis

- Univariate and bivariate analysis of mood, energy, productivity, and word count
- Correlation analysis (Pearson and Spearman) across all numeric features
- Time-series analysis with 7-day rolling averages, monthly trends, and day-of-week patterns
- Reflection streak analysis — consistency, longest streak, current streak

NLP and Sentiment Analysis

- Word frequency, bigrams/trigrams, and TF-IDF to surface topically distinctive terms
- VADER sentiment analysis run on raw (uncleaned) text to preserve punctuation and context signal

Emotion Detection and Topic Modeling

- Transformer-based emotion classification (Hugging Face), explicitly documented as an NLP classification task, not a psychological assessment
- Latent Dirichlet Allocation (7 topics) to identify recurring themes across reflections

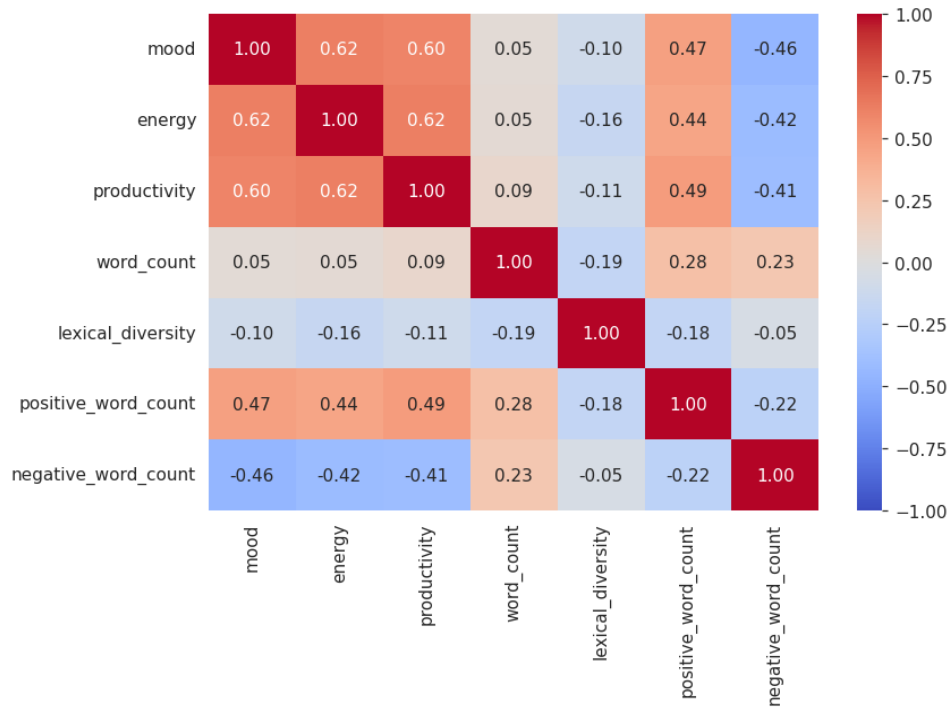
Machine Learning

- Target: next recorded reflection's productivity rating
- Chronological (not random) train/test split, to avoid lookahead leakage on time-dependent data
- Compared Linear Regression, Random Forest, and Gradient Boosting against a mean-prediction baseline

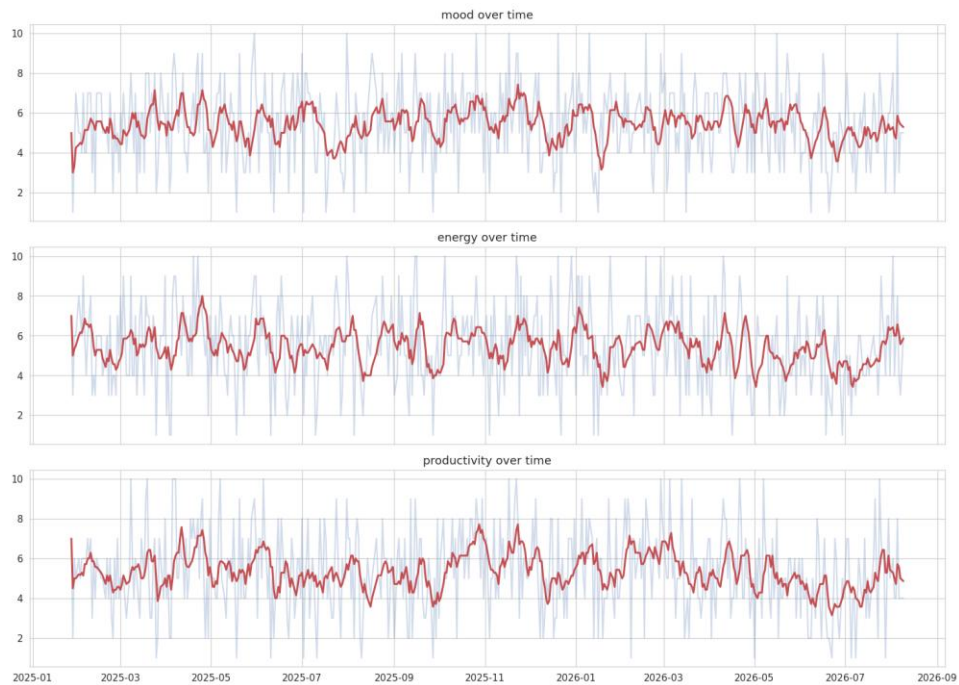
Key Findings

Metric Relationship	Finding
Mood vs. Energy vs. Productivity	Moderately-to-strongly correlated with each other ($r \approx 0.60\text{--}0.62$)
Positive word count vs. mood	$r = +0.47$ — text content agrees with numeric ratings
Negative word count vs. mood	$r = -0.46$
Productivity: weekday vs. weekend	Mild, real dip on weekends
Journaling consistency	92.3% of days written; longest streak 54 days
Next-day productivity prediction	No model meaningfully beat the mean baseline (honest negative result)

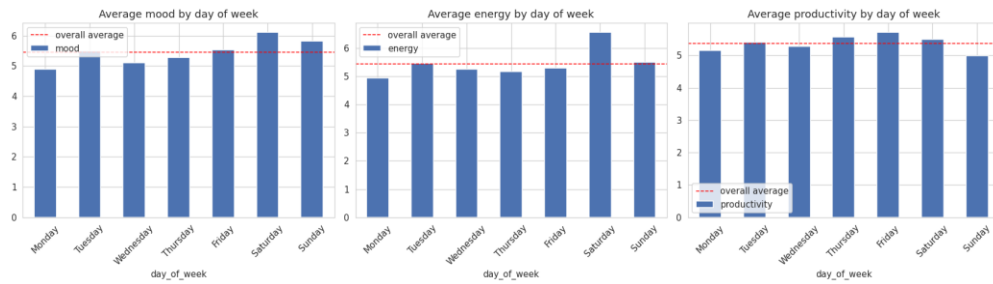
Correlation Between Metrics



Mood, Energy, and Productivity Over Time



Day-of-Week Patterns



Machine Learning Results

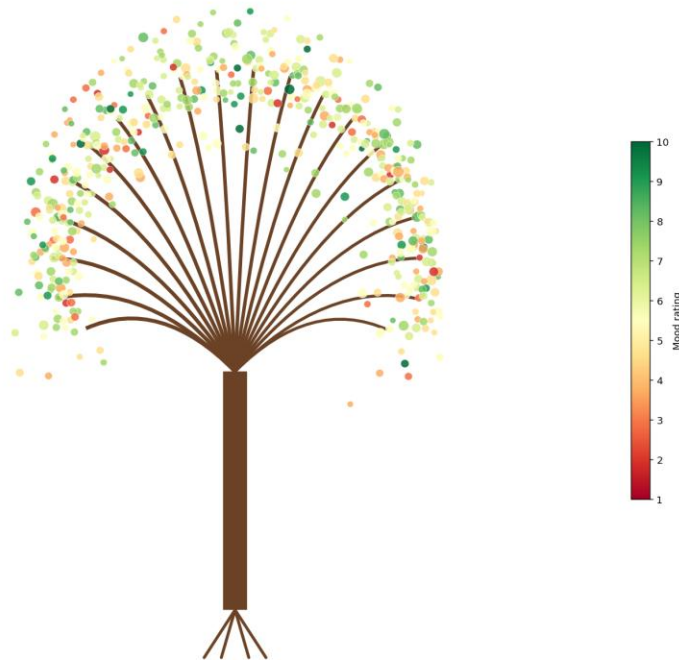
Model	MAE	RMSE	R ²
Linear Regression	1.644	2.029	-0.107
Random Forest	1.675	2.052	-0.132
Gradient Boosting	1.783	2.181	-0.279
Baseline (predict mean)	1.658	2.037	0.000

None of the three models produced a positive R^2 , meaning none explained more variance in next-day productivity than simply predicting the average. This is reported as an honest result rather than adjusted to look favorable: the dataset was deliberately built with only a mild autocorrelation under substantial noise, and roughly 500 records is a small sample for this kind of prediction task. A strong R^2 here would have been a red flag that something in the evaluation was wrong.

The Reflection Tree

The project's signature visualization maps every element to real data rather than using decoration: roots represent the beginning of the journaling journey, the trunk represents overall consistency, branches represent months, and leaves represent individual reflections — colored by mood and sized by reflection length.

Daily Reflection Tree
Each leaf = one reflection | Color = mood | Size = reflection length



Limitations

- Synthetic dataset — patterns reflect the generator’s design, not real human behavior
- ~500 records is small for machine learning; results should not be over-generalized
- Self-reported ratings are inherently subjective
- Correlation and feature importance describe association, never causation
- Emotion and sentiment models can misclassify nuanced or sarcastic text

Conclusion

This project demonstrates a complete, honestly-executed data science lifecycle on a genuinely unstructured-to-structured problem: cleaning, statistically grounded EDA, NLP, sentiment and emotion analysis, topic modeling, and machine learning with proper time-aware evaluation. Mood, energy, and productivity move together in a defensible way, and the reflection text genuinely agrees with the numeric self-ratings. The machine learning section’s honest negative result is itself a meaningful finding — not every dataset supports reliable day-ahead prediction, and reporting that clearly is part of doing the analysis correctly.

Future Improvements

- Real-world dataset, collected with consent and full anonymization
- Larger sample size for more reliable machine learning
- Multilingual sentiment analysis
- More advanced topic modeling (e.g. BERTopic)
- Explainable AI (SHAP) for model interpretability